

SECOND ORDER ODES

66

K.V.B.

$$A \frac{d^2 y}{dt^2} + B \frac{dy}{dt} + Cy = 0$$

A, B, C

Constants
(for now!)

No forcing
term $\Rightarrow$
Null solutions

No forcing
term
(for now!)

$$\frac{d^2 y}{dt^2}$$

y''

Second derivative

Acceleration term

Bending term in a
Taylor series

[change in
slope]

Second order

$\Rightarrow$ Two null solutions

$$y_h = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

$\rightarrow$ Constant coefficients

A, B, C leads to
exponential solutions

$\rightarrow$ s_1 & s_2 may be the
same is a special
case

$\rightarrow$ Two arbitrary constants

$\rightarrow$ C_1 & C_2

Two arbitrary constants (C_1 & C_2)

$\Rightarrow$ Two initial conditions

$$y(0) \text{ \& \& } y'(0)$$

"velocity"
or "slope"

EXAMPLE OF HARMONIC MOTION

Basic Equation of Motion
in Physics & Engineering

$$A \frac{d^2 y}{dt^2} + B \frac{dy}{dt} + Cy = 0$$

$\rightarrow [B=0]$

$$A \frac{d^2 y}{dt^2} + Cy = 0$$

← It's actually
the Newton's Law!

$\rightarrow [A = \text{mass}(m)]$

$\rightarrow \{B = \text{damping term}\}$
currently zero

$$m \cdot y'' + ky = 0$$

Two initial
Conditions:

$y(0) \rightarrow$ Initial
position

$y'(0) \rightarrow$ Initial
velocity

mass $\times$ accele-
ration
($\frac{d^2 y}{dt^2}$)

$$[F=ma]$$

$y \rightarrow$ position of
an object in
motion

$k \rightarrow$ restoring
factor
acting opposite
to the direction
of y

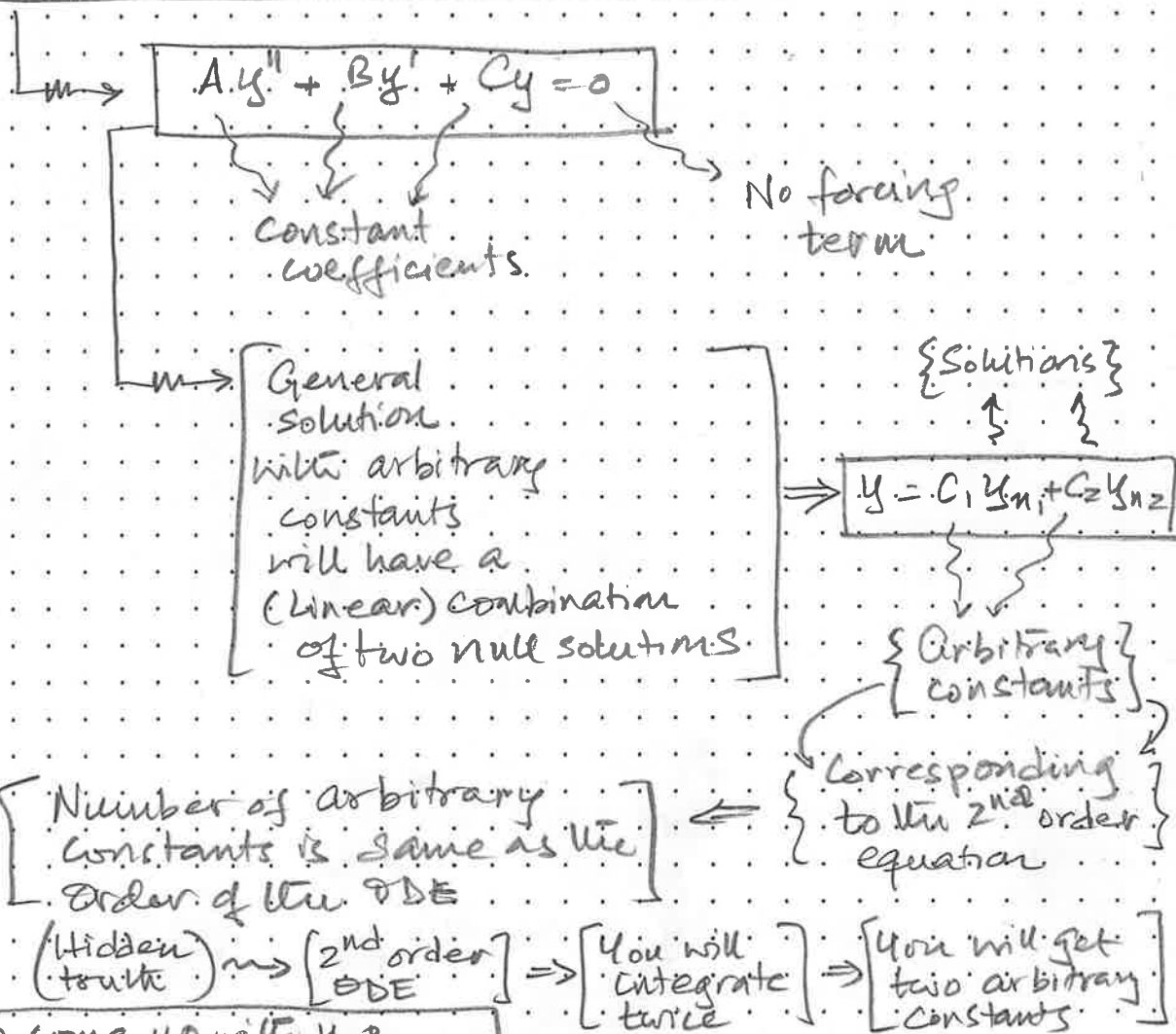
Application: 1. Spring with some mass going up & down
2. Pendulum going back & forth

Suppose: $m=1$ & $k=1$ we have second order ODE

$$y'' + y = 0$$

$\rightarrow$ Second derivative
plus the function
is zero

Let's go back to the basic form
of the 2nd order, linear, homogenous
ODE with constant coefficients



To come up with y :

Find: Two solutions y_{n1} & y_{n2}

Multiply: y_{n1} & y_{n2} with same arbitrary constants (C_1 & C_2)

Add: $(C_1 y_{n1} + C_2 y_{n2})$ or (take the linear combination of the two terms)

Conditions:

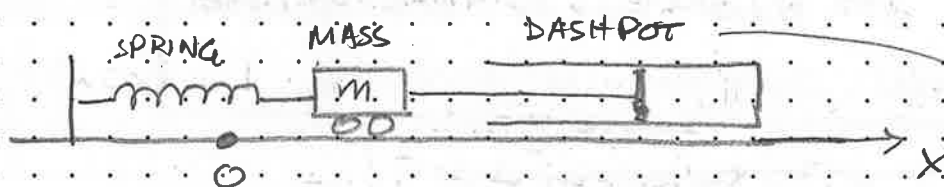
Linear independence

⊛ y_{n1} cannot be a multiple of y_{n2}
 $[y_{n1} \neq m y_{n2}]$

⊛ Initial conditions
 satisfy the C_1 & C_2 requirement
 $[y(0), y'(0)]$

EXAMPLE OF SPRING-MASS-DASHPOT SYSTEM

KVB



equilibrium point
where nothing happens
(that's where the mass wants to be)

x is Dependent variable (distance)

t is Independent variable (time)

eg: Dashpot does not allow the door to be slammed shut.

$$m x'' = -kx - cx'$$

$\{ \text{Force} \}$	$=$	$\{ \text{Spring force} \}$	$\{ \text{Dashpot damping force} \}$
$[\text{Newton's Law}]$		$\{ \text{opposes the motion} \}$	$\{ \text{opposes velocity} \}$
		$[\text{Hooke's Law}]$	$[?]$

$$m x'' + c x' + k x = 0$$

Divide by mass (m)

$$x'' + \frac{c}{m} x' + \frac{k}{m} x = 0$$

$\frac{c}{m}$ Damping constant (per unit mass)
 $\frac{k}{m}$ Spring constant (per unit mass)

To solve the ODE:

Find: Two linearly independent solutions (y_1, y_2)

Basic Method :

- ① Let $x = e^{st}$ where s is some constant.
- ② Substitute in the 2nd order, linear, homogeneous equation

$$x'' + Ax' + Bx = 0$$

$$\begin{cases} x'' = s^2 e^{st} \\ x' = s e^{st} \\ x = e^{st} \end{cases}$$

$$\Rightarrow s^2 e^{st} + s A e^{st} + B e^{st} = 0$$

$$\Rightarrow s^2 + As + B = 0 \quad \rightarrow \quad \begin{cases} \text{This is a} \\ \text{"characteristic"} \\ \text{equation."} \\ \text{with two roots.} \end{cases}$$

Possibilities for roots (s_1 & s_2):

- (i) $s_1 \neq s_2$ $\rightarrow$ Real & distinct $\rightarrow$ Easy to handle
- (ii) $s_1 \neq s_2$ $\rightarrow$ Complex conjugates $\rightarrow$ Hard to handle
- (iii) $s_1 \neq s_2$ $\rightarrow$ Real & equal $\rightarrow$ Puzzling to handle

EXAMPLE PROBLEM WITH REAL & DISTINCT ROOTS

71

KWB

Solve

$$y'' + 4y' + 3y = 0$$

with initial states

$$[y(0) = 1]$$

$$[y'(0) = 0]$$

Interpret this problem as a spring-mass-dashpot system. (Optional but could be insightful!)



{ Damping constant (per unit mass) } $\rightarrow 4$

{ Spring constant (per unit mass) } $\rightarrow 3$

{ Initial state } $\rightarrow$ I'm pulling the mass to a location at time $t=0$
 $y(0) = 1$

{ Initial state } $\rightarrow$ The velocity at that location is zero.
 $y'(0) = 0$
 $\Rightarrow$ I'm just letting the mass go without giving it any velocity at the stretched location

Solution

Characteristic equation of the ODE:

$$y'' + 4y' + 3y = 0$$

$$s^2 + 4s + 3 = 0$$

$$\Rightarrow (s+3)(s+1) = 0$$

[Real & distinct roots]

$$s_1 = -3$$

$$s_2 = -1$$

① Compare this ODE equation to

$$x'' + Ax' + Bx = 0$$

② Let $x = e^{st}$

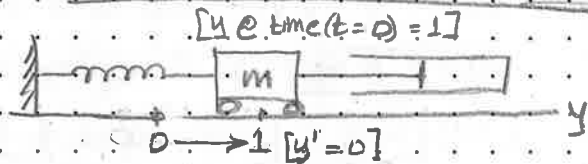
$$s^2 e^{st} + 3A e^{st} + 3B e^{st} = 0$$

③ Characteristic equation is

$$s^2 + As + B = 0$$

$\hookrightarrow$ BTW, you don't need to do this!

$\hookrightarrow$ You can just write the characteristic equation for your problem, directly!



Recall: $y = C_1 y_{n1} + C_2 y_{n2}$
 $= C_1 e^{s_1 t} + C_2 e^{s_2 t}$

Where now $s_1 = -3$ & $s_2 = -1$

$\therefore y = C_1 e^{-3t} + C_2 e^{-t}$

Apply initial states: $[y'(0) = 0]$ & $[y(0) = 1]$

$y'(0) = 0 = -3C_1 e^{-3(0)} - C_2 e^{-1(0)}$

$\Rightarrow -3C_1 - C_2 = 0$

$y(0) = 1 = C_1 e^{-3(0)} + C_2 e^{-1(0)}$

$\Rightarrow C_1 + C_2 = 1$

We ended up with a pair of simultaneous linear equations



$C_1 + C_2 = 1$
 $-3C_1 - C_2 = 0$

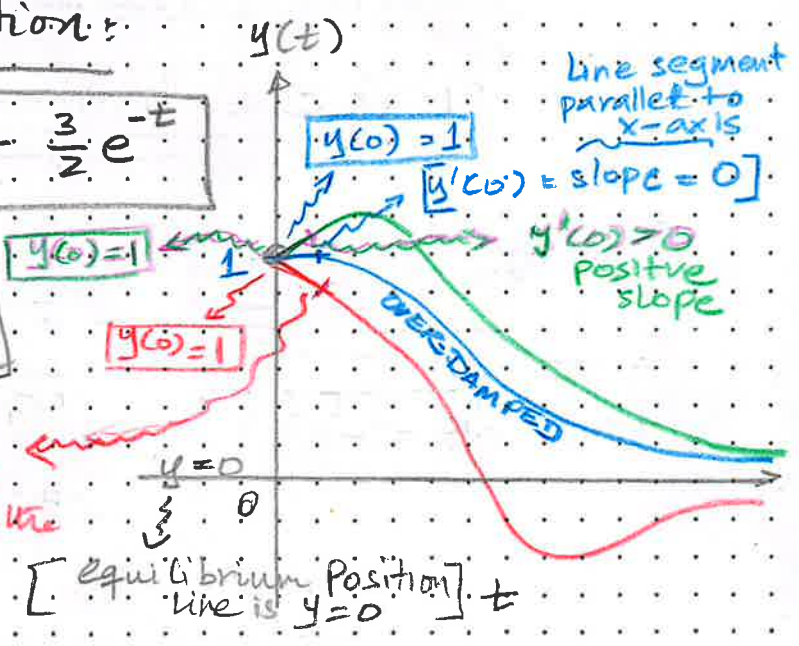
Adding: $-2C_1 = 1 \Rightarrow C_1 = -\frac{1}{2}$ & $C_2 = \frac{3}{2}$

So we have our solution:

$y = -\frac{1}{2} e^{-3t} + \frac{3}{2} e^{-t}$

Ⓢ All solution curves asymptote with $y=0$ the equilibrium position line.

$y'(0) < 0$ strongly negative slope
 "push" on the mass



EXAMPLE WITH COMPLEX CONJUGATE ROOTS

Solve

$$y'' + 4y' + 5y = 0$$

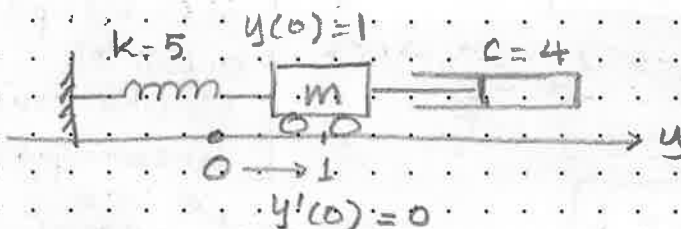
with initial states

$$[y(0) = 1] \text{ \& } [y'(0) = 0]$$

(Optional) physical interpretation with
spring-mass-damping system:

{ Damping constant } $\rightarrow 4$ { same as previous }
per unit mass. { example }

{ Spring constant } $\rightarrow 5$ { much higher }
per unit mass. { than previous }
{ example }



(Spring is pulling a little harder than the previous case)
 $\Rightarrow$ Stiffer spring!

[But the spring is stiffer. $\therefore$ The mass is getting pulled back harder!]

Characteristic equation:

$$y'' + 4y' + 5y = 0 \rightarrow s^2 + 4s + 5 = 0$$

using the Quadratic roots formula

$$s = \frac{-4 \pm \sqrt{4^2 - 4(1)(5)}}{2(1)}$$

$$\Rightarrow s = \frac{-4 \pm \sqrt{-4}}{2} = -2 \pm \frac{2\sqrt{-1}}{2}$$

$$\Rightarrow s = -2 \pm i$$

$$s_1 = -2 + i$$

$$s_2 = -2 - i$$

Recall: $y = C_1 e^{s_1 t} + C_2 e^{s_2 t}$

$y = C_1 e^{(2+i)t} + C_2 e^{(-2-i)t}$

$\Rightarrow y = C_1 e^{-2t} e^{it} + C_2 e^{-2t} e^{-it}$

$= e^{-2t} [C_1 e^{it} + C_2 e^{-it}]$

These terms can be expanded using Euler's formulas.
 $e^{i\theta} = \cos\theta + i\sin\theta$
 $e^{-i\theta} = \cos\theta - i\sin\theta$

$= e^{-2t} [C_1 (\cos t + i\sin t) + C_2 (\cos t - i\sin t)]$

$y = e^{-2t} [(C_1 + C_2) \cos t + i(C_1 - C_2) \sin t]$

$\Rightarrow y = (C_1 + C_2) e^{-2t} \cos t + i(C_1 - C_2) e^{-2t} \sin t$

Theorem: If $u + iv$ is the solution to a real ODE (linear, homogenous, have constant coefficients) i.e. $y'' + Ay' + By = 0$ then, u, v are real solutions;

Proof: $(u + iv)'' + A(u + iv)' + B(u + iv) = 0$

$\Rightarrow (u'' + Au' + Bu) + i(v'' + Av' + Bv) = 0$

Real part Imaginary part

The only this is possible is if real part = 0 & imaginary part = 0 separately

$\Rightarrow u'' + Au' + Bu = 0 \Rightarrow u$ is a solution

$v'' + Av' + Bv = 0 \Rightarrow v$ is a solution! ✓

We know that

$$y = \underbrace{(C_1 + C_2) e^{-2t} \cos t}_u + i \underbrace{(C_1 - C_2) e^{-2t} \sin t}_v$$

From the theorem we proved earlier,

the solutions are the linear combination of $e^{-2t} \cos t$ & $e^{-2t} \sin t$

$$y = (C_1 + C_2) e^{-2t} \cos t + (C_1 - C_2) e^{-2t} \sin t$$

Note: C_1 & C_2 are arbitrary constants

Notice that "i" is dropped following the theorem

$$\Rightarrow y = C_1 e^{-2t} \cos t + C_2 e^{-2t} \sin t$$

$$y = e^{-2t} (C_1 \cos t + C_2 \sin t)$$

Exponential term that decays the amplitude

combination of two sinusoidal oscillations with different amplitudes (C_1 & C_2) but with the same frequency

Apply initial states: $[y(0) = 1]$ & $[y'(0) = 0]$

$$\textcircled{1} \underline{y(0) = 1} = \underbrace{e^{-2(0)}}_1 [\underbrace{C_1 \cos(0)}_1 + \underbrace{C_2 \sin(0)}_0] \Rightarrow \boxed{C_1 = 1}$$

② $y'(0) = 0$

$$\Rightarrow y'(t) = \frac{d}{dt} [e^{-2t} (C_1 \cos t + C_2 \sin t)]$$

$$\Rightarrow y'(t) = \frac{d}{dt} [e^{-2t} (C_1 \cos t + C_2 \sin t)]$$

Use product rule

$$[(uv)'] = u'v + uv'$$

Here:

$$u' = \frac{d}{dt} (e^{-2t}) = -2e^{-2t}$$

$$v' = \frac{d}{dt} (C_1 \cos t + C_2 \sin t) = -C_1 \sin t + C_2 \cos t$$

Substituting $(uv)'$ with $t=0$ we have

$$y'(0) = [-2e^{-2(0)}] (C_1 \cos(0) + C_2 \sin(0)) + [e^{-2(0)}] (-C_1 \sin(0) + C_2 \cos(0)) = 0$$

$$\Rightarrow y'(0) = (-2)(1) + C_2(1) = 0 \Rightarrow \boxed{C_2 = 2}$$

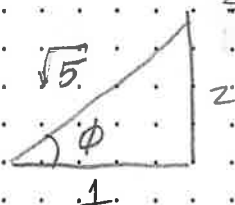
∴ The solution:

$$y = e^{-2t} (\cos t + 2 \sin t)$$

Further, we can use some trig to rewrite the solution...

↳ Amplitude of $\cos t$ is 1

↳ Amplitude of $\sin t$ is 2



$$y = \sqrt{5} e^{-2t} \cos(t - \phi)$$

where

$\phi \approx 1.1 \text{ rad}$

or
 63.4°

Aside: $R \cos(t-\phi) = (R \cos \phi) \cos t + (R \sin \phi) \sin t$
 compare R.H.S with $(\cos t + 2 \sin t)$

$$\Rightarrow R \cos \phi = 1 \quad \& \quad R \sin \phi = 2$$

$$\therefore R^2 \cos^2 \phi + R^2 \sin^2 \phi = 1^2 + 2^2$$

$$\Rightarrow R^2 (\underbrace{\cos^2 \phi + \sin^2 \phi}_1) = 5$$

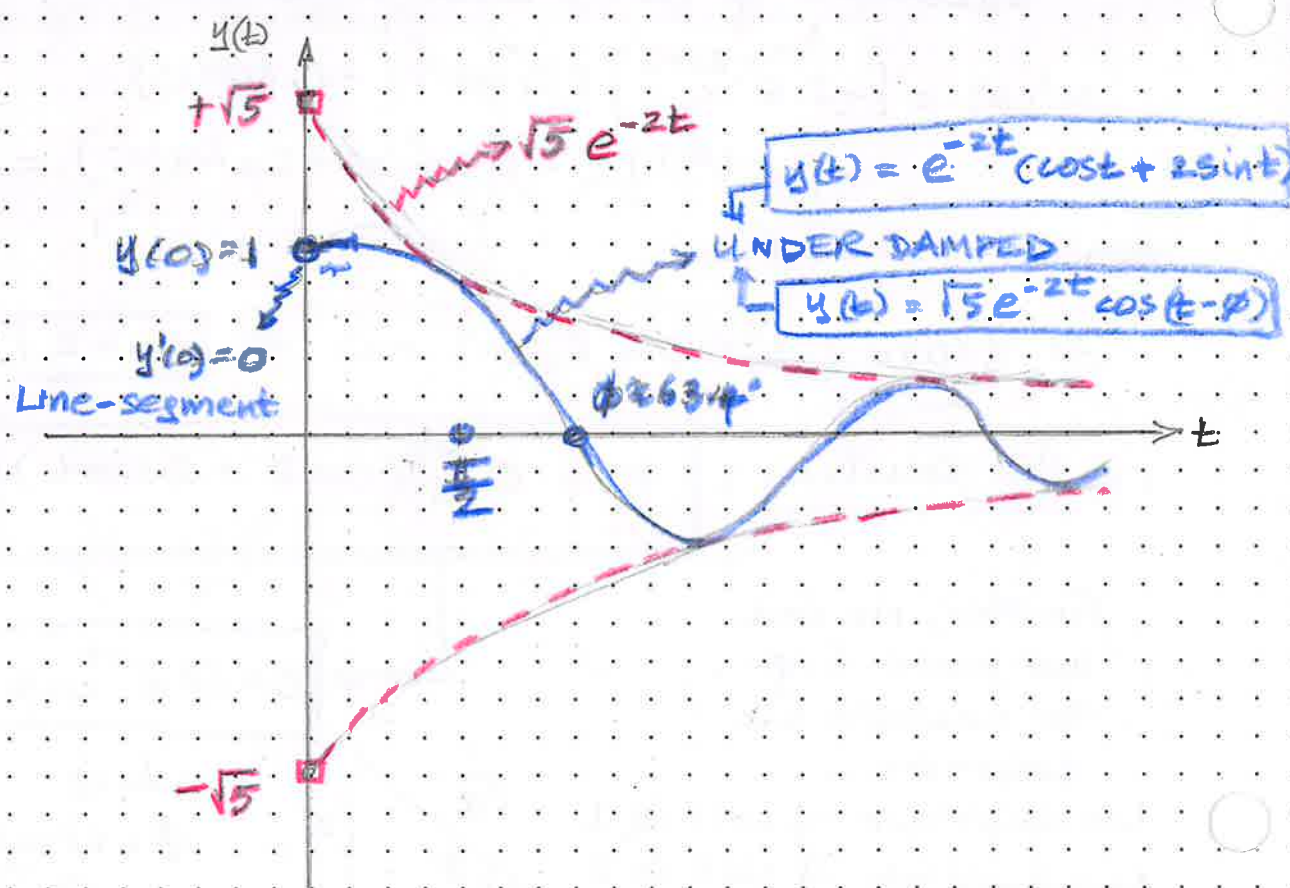
$$\Rightarrow R = \sqrt{5}$$

Also, $\frac{R \sin \phi}{R \cos \phi} = \frac{2}{1} \Rightarrow \tan \phi = 2$

$$\Rightarrow \phi = \tan^{-1}(2)$$

$$\Rightarrow \phi \approx 63.4^\circ \text{ or } 1.107 \text{ rad}$$

$$\cos t + 2 \sin t = \sqrt{5} \cos(t - 1.1)$$



INTUITIVE EXAMPLE WITH EXPLANATION

73

$$y'' = -y$$

Suppose

$$m=1$$

K.V.B.

$$k=1$$

without damping

Let's guess a solution

$$[\text{We have } y'' + y = 0]$$

Second derivative of a sine function is a negative sine function

$$\text{So, } \frac{d^2 \sin(t)}{dt^2} = -\sin(t) = y'' \quad \boxed{\text{R.H.S.}}$$

$$\text{So, } -y = -\sin(t) \quad \boxed{\text{L.H.S.}}$$

So, I am thinking the solution could have some sines & cosines

Now, let's try and extend our ideas to solving

$$m \cdot y'' + k \cdot y' = 0$$

THIS IS AN UNDAMPED ODE TYPE

I expect my null solution to have sines & cosines

Homogeneous equation
i.e. no forcing term

Recall: I need two null solutions

$$y_n = C_1 \cos(\text{something}) + C_2 \sin(\text{something})$$

What is this "something"?

"Something" could include k , & m

Try:

$$y_{n1} = C_1 \cos\left(\sqrt{\frac{k}{m}} t\right)$$

$$m y_{n1}'' + k y_{n1} = 0$$

$$\Rightarrow -\left(\frac{k}{m}\right) C_1 \cos\left(\sqrt{\frac{k}{m}} t\right)$$

$$+ (k) C_1 \cos\left(\sqrt{\frac{k}{m}} t\right) = 0$$

Similarly:

$$y_{n2} = C_2 \sin\left(\sqrt{\frac{k}{m}} t\right)$$

$$m y_{n2}'' + k y_{n2} = 0$$

$$\Rightarrow -\left(\frac{k}{m}\right) C_2 \sin\left(\sqrt{\frac{k}{m}} t\right) + (k) C_2 \sin\left(\sqrt{\frac{k}{m}} t\right)$$

$$= 0 \quad \checkmark$$

$$y_n = C_1 \cos\left(\sqrt{\frac{k}{m}} t\right) + C_2 \sin\left(\sqrt{\frac{k}{m}} t\right)$$

$\rightarrow$ Has two null solutions that independent
 $\rightarrow$ free harmonic motion

But notice that this solution is not the same as $y_n = C_1 e^{s_1 t} + C_2 e^{s_2 t}$, that we originally envisioned as a solution to homogeneous linear, second order ODE with constant coefficients.

- ⊕ Cosines and sines are related to exponentials
- ⊕ We can use the Euler's formula that involves complex exponentials.

$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$$

Let's introduce " ω_n " for natural motion
 natural frequency of motion

Say: $\omega_n = \sqrt{\frac{k}{m}}$

$$m y'' + k y = 0$$

$$\Rightarrow y'' + \frac{k}{m} y = 0$$

$$\Rightarrow y'' + \omega_n^2 y = 0$$

⚡️ null solution

$$y_n = C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t)$$

FORMAL APPROACH AND EXPLANATION

80
KVB

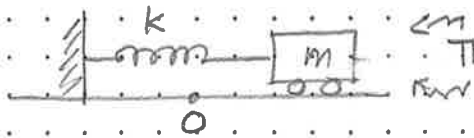
Now let's do this in a more formal way

$$m y'' + k y = 0$$

Second order ODE
homogenous,
linear,
UNDAMPED-TYPE.

$$\Rightarrow y'' + \frac{k}{m} y = 0$$

We define $\omega_n^2 = \frac{k}{m}$ where $\omega_n \rightarrow$ natural frequency of motion



This is how the system looks like.

Now to solve $[y'' + \frac{k}{m} y = 0]$ or $[y'' + \omega_n^2 y = 0]$

(I) Let $y_n = e^{st} \Rightarrow (y'' + \omega_n^2 y) = s^2 e^{st} + \omega_n^2 e^{st} = 0$

$\therefore s^2 e^{st} + \omega_n^2 e^{st} = 0 \Rightarrow s^2 + \omega_n^2 = 0 \Rightarrow s^2 = -\omega_n^2$

Characteristic equation

(II) Characteristic equation: $s^2 = -\omega_n^2$

$\Rightarrow s = \pm i \omega_n \Rightarrow \begin{cases} s_1 = i \omega_n \\ s_2 = -i \omega_n \end{cases}$

(III) General form of the solution:

$$y_n = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

$\therefore y_n = C_1 e^{i \omega_n t} + C_2 e^{-i \omega_n t}$

$\Rightarrow y_n = C_1 [\cos(\omega_n t) + i \sin(\omega_n t)] + C_2 [\cos(\omega_n t) - i \sin(\omega_n t)]$

$\Rightarrow y_n = (C_1 + C_2) \cos(\omega_n t) + i(C_1 - C_2) \sin(\omega_n t)$

Recall theorem: If $(u + iv)$ is the solution to a real 2nd order ODE that is linear, homogenous & has constant coefficients then u, v are real solutions.

$$y_n = C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t)$$

$$\left\{ \text{where } \omega_n = \sqrt{\frac{k}{m}} \right\}$$

→ We should figure out C_1 & C_2

→ Use Two initial conditions

1. $y(0)$ at $t=0$ → [Position of the spring or pendulum at $t=0$]

2. $y'(0)$ at $t=0$ → [Velocity of the spring or pendulum at $t=0$]

IV

Apply $y(0)$ at $t=0$ to $y_n = C_1 \cos(\omega_n t) + C_2 \sin(\omega_n t)$:

$$y = y(0) = C_1 \cos(0) + C_2 \sin(0)$$

$$\Rightarrow C_1 = y(0) \Rightarrow y_n = y(0) \cos(\omega_n t) + C_2 \sin(\omega_n t)$$

Apply $y'(0)$ at $t=0$ to $y_n = y(0) \cos(\omega_n t) + C_2 \sin(\omega_n t)$:

$$y'_n = \frac{dy_n}{dt} = -y(0) \omega_n \sin(\omega_n t) + C_2 \omega_n \cos(\omega_n t)$$

$$\Rightarrow y'(0) = -y(0) \omega_n \sin(0) + C_2 \omega_n \cos(0)$$

$$\Rightarrow y'(0) = C_2 \omega_n \Rightarrow C_2 = \frac{y'(0)}{\omega_n}$$

$$\Rightarrow y_n = y(0) \cos(\omega_n t) + \frac{y'(0)}{\omega_n} \sin(\omega_n t)$$

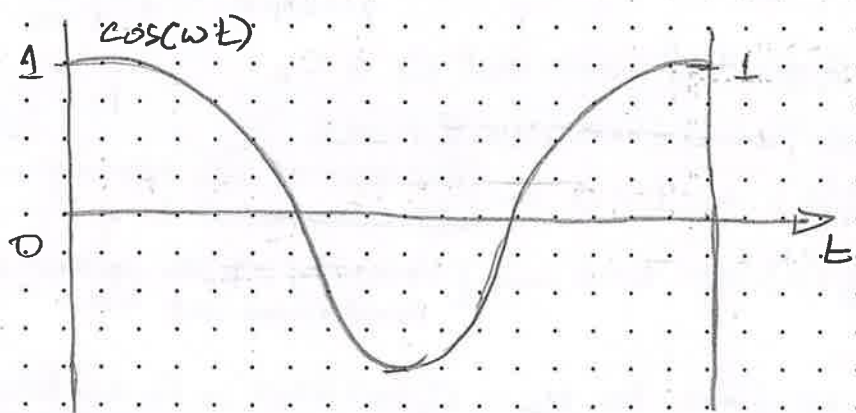
→ That tells me about the motion forever & ever

$$\left\{ \text{where } \omega_n = \sqrt{\frac{k}{m}} \right\}$$

→ That's the free harmonic motion, no damping.

→ We can write this using complex exponentials. But I am happy with this for now.

Let's draw the graph of $\cos(\omega t)$.



$$\omega T = 2\pi \quad \leftarrow \text{Angular frequency}$$

$$\Rightarrow \omega = \text{radians per second}$$

{ There is another frequency (f) measured in Hertz }

$$f \cdot T = 1$$

$$\Rightarrow f = \text{cycles per second}$$

$$\omega = 2\pi f$$

$$\Rightarrow \text{One cycle} = (2\pi) \text{ radians}$$

Let's (finally) rewrite

$$y_n = y(0) \cos(\omega_n t) + \frac{y'(0)}{\omega_n} \sin(\omega_n t)$$

in terms of exponentials:

$$y_n = c_1 e^{i\omega_n t} + c_2 e^{-i\omega_n t}$$